🔴 RED TEAM ENGAGEMENT REPORT

Prepared By: Yusuf Tahir Ajah
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Engagement Type: Simulated Red Team Assessment

1. EXECUTIVE SUMMARY:

This report presents the results of a simulated Red Team engagement conducted to evaluate the security posture of the target environment. The objective was to identify vulnerabilities across the attack lifecycle, from reconnaissance to data exfiltration, and demonstrate potential real-world exploitation scenarios.

During the engagement, multiple critical vulnerabilities were identified, including weak credential policies, successful phishing-based initial access, privilege escalation, and lateral movement across systems. Sensitive data was successfully accessed and exfiltrated, demonstrating significant risk to organizational assets.

The findings highlight gaps in security controls, monitoring, and user awareness. Immediate remediation is required to mitigate potential real-world attacks

2. METHODOLOGY:

The engagement followed a structured Red Team methodology simulating real-world adversarial behavior:

- Reconnaissance (OSINT & scanning)
- Initial Access (phishing/social engineering)
- Exploitation (vulnerability abuse)
- Lateral Movement (internal pivoting)
- Post-Exploitation (credential access)
- Exfiltration (data extraction)

3. FINDINGS:

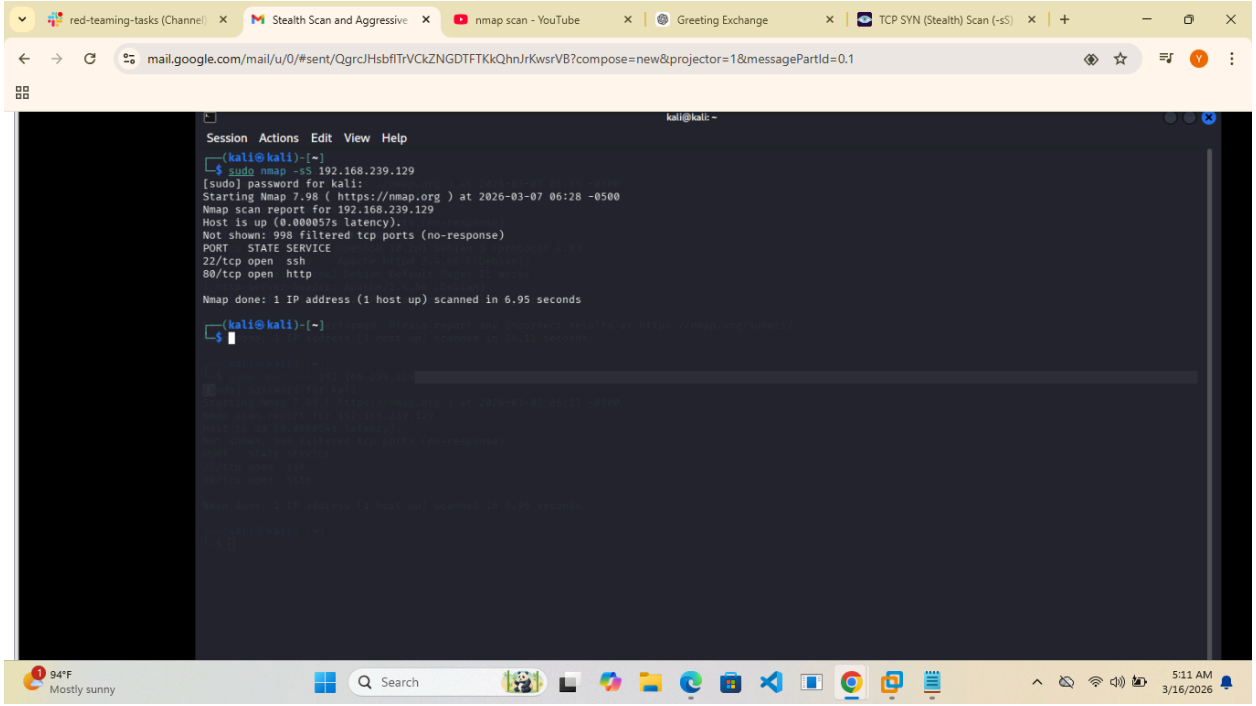
3.1 Finding 1: Open Ports and Service Exposure.

Severity: Medium

Description:
Network reconnaissance revealed multiple open ports and exposed services, increasing the attack surface.

Evidence:

- Open ports identified via scanning tools
- Running services detected



Impact:
Attackers can identify entry points and vulnerable services.

Recommendation:

- Close unused ports
- Implement firewall restrictions
- Regularly audit exposed services

3.2 Finding 2: Successful Phishing Attack:

Severity: High

Description:
A phishing simulation successfully captured user credentials, enabling unauthorized access.

Evidence:

- Login credentials captured
- Victim interaction with phishing page

Impact:
Unauthorized access to internal systems.

Recommendation:

- Conduct user awareness training
- Enable Multi-Factor Authentication (MFA)
- Deploy email filtering solutions

3.3 Finding 3: Credential Dumping:

Severity: Critical

Description:
Post-exploitation techniques allowed extraction of credentials from memory.

Impact:
Full system compromise and privilege escalation.

Recommendation:

- Restrict administrative privileges
- Enable credential protection mechanisms
- Monitor suspicious processes

3.4 Finding 4: Lateral Movement:

Severity: High

Description:
Compromised credentials were used to move laterally across systems within the network.

Impact:
Expansion of attacker control across network.

Recommendation:

- Network segmentation
- Monitor internal traffic
- Limit credential reuse

3.5 Finding 5: Data Exfiltration:

Severity: Critical

Description:
Sensitive data was successfully exfiltrated from the target system.

Impact:
Loss of confidential data and potential regulatory violations.

Recommendation:

- Implement Data Loss Prevention (DLP)
- Monitor outbound traffic
- Encrypt sensitive data

4. ATTACK FLOW DIAGRAM:

The diagram below illustrates the complete attack lifecycle executed during the engagement:

Reconnaissance → Initial Access → Exploitation → Lateral Movement → Exfiltration

5. RECOMMENDATIONS:

The following actions are recommended to improve the organization's security posture:

- Enforce Multi-Factor Authentication (MFA)
- Implement strong password policies
- Conduct regular security awareness training
- Patch and update systems regularly
- Deploy endpoint detection and response (EDR)
- Restrict administrative privileges
- Monitor logs and network activity

- Segment internal networks
- Implement Data Loss Prevention (DLP) solutions

6. CONCLUSION:

This Red Team engagement successfully demonstrated how an attacker can exploit multiple weaknesses to gain unauthorized access, escalate privileges, move laterally, and exfiltrate sensitive data.

The identified vulnerabilities pose significant risks if left unaddressed. By implementing the recommended security controls, the organization can significantly reduce its attack surface and improve its resilience against real-world cyber threats.